2.21 Differentiation: Standard Functions

Worked Solutions 1–37

1 a)

$\frac{dy}{dx} = 5(x+3)^4$.

1 b)

$\frac{dy}{dx} = 24(4x+1)^5$.

1 c)

$\frac{dy}{dx} = -20(6-5x)^3$.

1 d)

$\frac{dy}{dx} = 14(2x-3)^6$.

1 e)

$\frac{dy}{dx} = -(7-4x)^{-\frac{3}{4}}$.

1 f)

$y = (x+2)^{-1}$ so $\frac{dy}{dx} = -(x+2)^{-2}$.

1 g)

$y = 3(x-5)^{-2}$ so $\frac{dy}{dx} = -6(x-5)^{-3}$.

1 h)

$y = (8-3x)^{-4}$ so $\frac{dy}{dx} = 12(8-3x)^{-5}$.

1 i)

$y = (2+x)^{\frac{1}{2}}$ so $\frac{dy}{dx} = \frac{1}{2}(2+x)^{-\frac{1}{2}}$.

1 j)

$y = (3x-1)^{-\frac{1}{2}}$ so $\frac{dy}{dx} = -\frac{3}{2}(3x-1)^{-\frac{3}{2}}$.

2 a)

$f'(x) = 35(7x-1)^4$, $f''(x) = 980(7x-1)^3$.

2 b)

$f'(x) = -\frac{1}{2}(3-x)^{-\frac{1}{2}}$, $f''(x) = -\frac{1}{4}(3-x)^{-\frac{3}{2}}$.

2 c)

$f(x) = (2x+9)^{-\frac{1}{3}}$, $f'(x) = -\frac{2}{3}(2x+9)^{-\frac{4}{3}}$, $f''(x) = \frac{16}{9}(2x+9)^{-\frac{7}{3}}$.

3 a)

$\frac{dy}{dx} = 10(2x-7)^4$ stat pt when $\frac{dy}{dx} = 0$. $10(2x-7)^4 = 0$ so $x = \frac{7}{2} \Rightarrow (\frac{7}{2}, 0)$.

3 b)

$y = 4x + (x-1)^{-1}$. $\frac{dy}{dx} = 4 - (x-1)^{-2}$ stat pt when $\frac{dy}{dx} = 0$. $4 - \frac{1}{(x-1)^2} = 0 \Rightarrow \frac{1}{(x-1)^2} = 4 \Rightarrow \frac{1}{4} = (x-1)^2$. $\Rightarrow \pm\frac{1}{2} = x - 1$ so $x = \frac{1}{2}$ or $x = \frac{3}{2}$. So stat point at $(\frac{1}{2}, 0)$ and $(\frac{3}{2}, 8)$.

4 a)

$$\begin{aligned} f'(x) &= 12(3x+1)^3 \\ &= 12[1(3x)^3 + 3(3x)^2 + 3(3x) + 1] \\ &= 324x^3 + 324x^2 + 108x + 12 \end{aligned}$$

4 b)

$$\begin{aligned} f(x) &= 1(3x)^4 + 4(3x)^3 + 6(3x)^2 + 4(3x) + 1 \\ f(x) &= 81x^4 + 108x^3 + 54x^2 + 12x + 1 \\ f'(x) &= 324x^3 + 324x^2 + 108x + 12 \end{aligned}$$

5)

$\frac{dy}{dx} = \frac{1}{2}(x+7)^{-\frac{1}{2}}$ so $\left.\frac{dy}{dx}\right|_{x=2} = \frac{1}{2} \times \frac{1}{3} = \frac{1}{6}$. When $x = 2$, $y = 3$. $y - 3 = \frac{1}{6}(x-2) \Rightarrow 6y - 18 = x - 2$. So $6y - x = 16$.

6 a)

$\frac{dh}{dt} = 32(t-6)$ stat pt when $\frac{dh}{dt} = 0$. $32(t-6) = 0$ so $t = 6$, therefore 12.00 (noon). Minimum height is 200 m.

6 b)

$\frac{d^2h}{dt^2} = 32 > 0$ so it is a minimum.

7 a)

$\frac{dy}{dx} = 3\cos 3x$.

7 b)

$\frac{dy}{dx} = -5\sin 5x$.

7 c)

$\frac{dy}{dx} = -14\cos 2x$.

7 d)

$\frac{dy}{dx} = -3\sin 6x$.

7 e)

$\frac{dy}{dx} = \frac{1}{4}\sin\frac{1}{4}x.$

7 f)

$\frac{dy}{dx} = -2\cos 10x.$

7 g)

$\frac{dy}{dx} = \pi\sin\pi x.$

7 h)

$\frac{dy}{dx} = -\frac{1}{\pi}\sin\frac{x}{\pi}.$

8)

$\frac{dy}{dx} = 12\cos 6x$ so $\left.\frac{dy}{dx}\right|_{x=\pi} = 12\cos 6\pi = 12.$ At $x = \pi$, $y = 0$ so $y - 0 = 12(x - \pi)$. Therefore $y = 12x - 12\pi$.

9)

$\frac{dy}{dx} = -3\sin 3x$ so $\left.\frac{dy}{dx}\right|_{x=\frac{\pi}{2}} = -3\sin 3x = 3.$ At $x = \frac{\pi}{2}$, $y = 1$ so $y - 1 = 3(x - \frac{\pi}{2})$. Therefore $y = 3x + 1 - \frac{3}{2}\pi$.

10)

$\frac{dy}{dx} = -\pi\cos\pi x$ so $\left.\frac{dy}{dx}\right|_{x=2} = -\pi.$ At $x = 2$, $y = 3$ so $y - 3 = \frac{1}{\pi}(x - 2)$. Therefore $y = \frac{1}{\pi}x + 3 - \frac{2}{\pi}$.

11)

$\frac{dy}{dx} = -3\sin 3x$ so $\frac{d^2y}{dx^2} = -9\cos 3x = -9y.$

12)

$\frac{dy}{dx} = -\sin x - 1$ stat pt when $\frac{dy}{dx} = 0.$ $-\sin x - 1 = 0 \Rightarrow \sin x = -1.$ $\Rightarrow x = \frac{3\pi}{2}$ for $0 < x < 2\pi$, therefore $(\frac{3\pi}{2}, -\frac{3\pi}{2})$.

13 a)

$p = 300 - 40\cos(0) = 260.$

13 b)

$\frac{dp}{dt} = \frac{20\pi}{3}\sin(\frac{\pi}{6}t).$

13 c)

$\left.\frac{dp}{dt}\right|_{t=6} = 0$ so the attraction has either a maximum or minimum amount of visitors.

13 d)

$\frac{d^2p}{dt^2} = \frac{10\pi^2}{9}\cos(\frac{\pi}{6}t)$ so $\left.\frac{d^2p}{dt^2}\right|_{t=6} = -\frac{10\pi^2}{9} < 0$ so it is a maximum amount of visitors.

14 a)

$\frac{ds}{dt} = -\pi \sin(\frac{\pi}{3}t)$.

14 b)

Max value of $\frac{ds}{dt}$ when $\sin(\frac{\pi}{3}t) = -1$ so $\frac{ds}{dt} = \pi$. Which is when $\frac{\pi}{3}t = \frac{3\pi}{2} \Rightarrow t = \frac{9}{2}$. It is a point of inflection, so $\frac{ds}{dt}$ is at its most steep. I.e. Acceleration is maximum.

15)

Within $0 < x < \frac{\pi}{3}$ the curve has two points where the gradient is the same, therefore the normal to the curve having the same gradient, so two solutions are expected. $\frac{dy}{dx} = -12\pi \sin(2\pi x)$. So $-12\pi \sin(2\pi x) = -6\pi \Rightarrow \sin(2\pi x) = \frac{1}{2}$. So $2\pi x = \frac{\pi}{6} \Rightarrow x = \frac{1}{12}$. Or $2\pi x = \frac{5\pi}{6} \Rightarrow x = \frac{5}{12}$. Therefore $x = \frac{1}{12}$ or $x = \frac{5}{12}$.

16)

$y = \sin(x^\circ) = \sin(\frac{\pi}{180}x)$. So $\frac{dy}{dx} = \frac{\pi}{180}\cos(\frac{\pi}{180}x) = \frac{\pi}{180}\cos(x^\circ)$.

17 a)

$\frac{dy}{dx} = \frac{1}{x}$.

17 b)

$\frac{dy}{dx} = \ln 3 \times 3^x$.

17 c)

$\frac{dy}{dx} = 5e^{5x}$.

17 d)

$\frac{dy}{dx} = \frac{2}{x}$.

17 e)

$\frac{dy}{dx} = -\ln 6 \times 6^x$.

17 f)

$y = e^{2x} - 2e^x + 1$ so $\frac{dy}{dx} = 2e^{2x} - 2e^x$.

18 a)

$\frac{dy}{dx} = 2e^x - 4$ stat pt when $\frac{dy}{dx} = 0$. So $2e^x - 4 = 0 \Rightarrow e^x = 2 \Rightarrow x = \ln 2$. Stat pt at $(\ln 2, 4 - 4\ln 2)$. $\frac{d^2y}{dx^2} = 2e^x$. $\left.\frac{d^2y}{dx^2}\right|_{x=\ln 2} = 2e^{\ln 2} = 4 > 0$ so it is a minimum.

18 b)

$\frac{dy}{dx} = \frac{1}{x} - 8$ stat pt when $\frac{dy}{dx} = 0$. So $\frac{1}{x} - 8 = 0 \Rightarrow x = \frac{1}{8}$. Stat pt at $(\frac{1}{8}, \ln\frac{1}{8} - 1)$. $\frac{d^2y}{dx^2} = -\frac{1}{x^2}$. $\left.\frac{d^2y}{dx^2}\right|_{x=\frac{1}{8}} = -\frac{1}{(\frac{1}{8})^2} = -64 < 0$ so it is a maximum.

18 c)

$\frac{dy}{dx} = 4x + 19 - \frac{5}{x}$ stat pt when $\frac{dy}{dx} = 0$. So $4x + 19 - \frac{5}{x} = 0 \Rightarrow 4x^2 + 19x - 5 = 0 \Rightarrow x = -5$ or $x = \frac{1}{4}$ but $x > 0$ so stat pt at $(\frac{1}{4}, \frac{39}{8} - 5\ln\frac{1}{4})$. $\frac{d^2y}{dx^2} = 4 + \frac{5}{x^2}$. $\left.\frac{d^2y}{dx^2}\right|_{x=\frac{1}{4}} = 4 + \frac{5}{(\frac{1}{4})^2} = 84 > 0$ so it is a minimum.

19 a)

$f'(x) = \frac{1}{x}$ so $\frac{1}{x} = 1 \Rightarrow x = 1$.

19 b)

$f'(x) = 2e^{2x}$ so $2e^{2x} = 1 \Rightarrow e^{2x} = \frac{1}{2} \Rightarrow 2x = \ln(\frac{1}{2})$. $\Rightarrow 2x = -\ln(2) \Rightarrow x = -\frac{1}{2}\ln(2)$.

19 c)

$f'(x) = \ln 3 \times 3^x$ so $\ln 3 \times 3^x = 1 \Rightarrow 3^x = \frac{1}{\ln 3}$. $\Rightarrow x = \log_3 \frac{1}{\ln 3}$.

20)

$\frac{dy}{dx} = 8x^{-1} - 10x$, $\frac{d^2y}{dx^2} = -\frac{8}{x^2} - 10 < 0$ so it is concave.

21)

$\frac{dy}{dx} = \frac{1}{x}$ so $\left.\frac{dy}{dx}\right|_{x=\frac{e}{5}} = \frac{5}{e}$ at $x = \frac{e}{5}$, $y = 1$.

$$\begin{aligned} y - 1 = \tfrac{5}{e}(x - \tfrac{e}{5}) &\Rightarrow ey - e = 5x - e \\ &\Rightarrow y = \tfrac{5}{e}x \end{aligned}$$

22)

$\frac{dy}{dx} = \ln 5 \times 5^x$ so $\left.\frac{dy}{dx}\right|_{x=2} = 25\ln 5$. At $x = 2$, $y = 25$.

$$\begin{aligned} y - 25 &= -\tfrac{1}{25\ln 5}(x - 2) \\ &\Rightarrow 25\ln 5\, y - 625\ln 5 = -x + 2 \\ &\Rightarrow 25\ln 5\, y + x = 2 + 625\ln 5 \end{aligned}$$

23 a)

$50 = 500 \times 20^{-\frac{1}{2}t} \Rightarrow -\frac{1}{2}t = \log_{20}\frac{1}{10}$. $\Rightarrow t = -2\log_{20}\frac{1}{10} = 1.537\ldots$ days. So 37 hours.

23 b)

$\frac{dP}{dt} = -250\ln 20 \times 20^{-\frac{1}{2}t}$. $\left.\frac{dP}{dt}\right|_{t=7} = -0.0209$ bugs per day.

24)

$\frac{dy}{dx} = \ln 9 \times 9^x - \ln 3 \times 3^x$ stat pt when $\frac{dy}{dx} = 0$.

$$\begin{aligned} &\text{So } \ln 9 \times 9^x - \ln 3 \times 3^x = 0 \\ &\Rightarrow \ln 3^2 \times (3)^{2x} - \ln 3 \times 3^x = 0 \\ &\Rightarrow 2\ln 3 \times (3)^{2x} - \ln 3 \times 3^x = 0 \\ &\Rightarrow 3^x \ln 3(2(3)^x - 1) = 0 \text{ but } 3^x \neq 0 \end{aligned}$$

$$\Rightarrow 2(3)^x - 1 = 0$$
$$\Rightarrow 3^x = \tfrac{1}{2} \Rightarrow x = \log_3 \tfrac{1}{2}$$

When $x = \log_3 \frac{1}{2}$, $y = \frac{1}{4} - \frac{1}{2} = -\frac{1}{4}$. $\frac{d^2y}{dx^2} = (\ln 9)^2 \times 9^x - (\ln 3)^2 \times 3^x$ so $\left.\frac{d^2y}{dx^2}\right|_{x=\log_3 \frac{1}{2}} > 0$ so a minimum.

25)

$\frac{dy}{dx} = 2\ln 2 \times 2^x$ so $\left.\frac{dy}{dx}\right|_{x=1} = 4\ln 2$. At $x = 1$, $y = 4$. Tangent line: $y - 4 = 4\ln 2(x-1)$. When $y = 0$, $-4 = 4\ln 2(x-1)$. $\Rightarrow -\frac{1}{\ln 2} = x - 1 \Rightarrow x = 1 - \frac{1}{\ln 2}$. Normal line: $y - 4 = -\frac{1}{4\ln 2}(x-1)$. When $y = 0$, $-4 = -\frac{1}{4\ln 2}(x-1)$. $\Rightarrow 16\ln 2 = x - 1 \Rightarrow x = 16\ln 2 + 1$. Area of triangle:

$$\tfrac{1}{2} \times [(16\ln 2 + 1) - (1 - \tfrac{1}{\ln 2})] \times 4 = \frac{2}{\ln 2} + 32\ln 2$$

26)

$\frac{dy}{dx} = \frac{1}{x}$ so $\left.\frac{dy}{dx}\right|_{x=e^b} = \frac{1}{e^b}$ at $x = e^b$, $y = b + a$.

$$y - (b+a) = \tfrac{1}{e^b}(x - e^b)$$
$$\Rightarrow e^b y - e^b(b+a) = x - e^b$$
$$\Rightarrow e^b y - x = -e^b + be^b + ae^b$$
$$\Rightarrow e^b y - x = (a + b - 1)e^b$$

27 a)

$\frac{dy}{dx} = 3\sec^2 3x$.

27 b)

$\frac{dy}{dx} = 8\sec^2 4x$.

27 c)

$\frac{dy}{dx} = -3\sec^2 9x$.

27 d)

$\frac{dy}{dx} = -\pi\sec^2 \pi x$.

27 e)

$\frac{dy}{dx} = -\frac{1}{2\pi}\sec^2(\frac{6}{\pi}x)$.

28)

$\frac{dy}{dx} = \frac{2\pi}{9}\sec^2(\frac{\pi}{9}x)$, $\left.\frac{dy}{dx}\right|_{x=\frac{9}{4}} = \frac{2\pi}{9} \times \frac{1}{\cos^2(\frac{\pi}{9}x)} = \frac{4\pi}{9}$. $x = \frac{9}{4}$, $y = 4 + 2\tan(\frac{\pi}{9} \times \frac{9}{4}) = 6$.

$$y - 6 = \tfrac{4\pi}{9}(x - \tfrac{9}{4})$$
$$\Rightarrow 9y - 54 = 4\pi x - 9\pi$$
$$\Rightarrow y = \tfrac{4\pi}{9}x + 6 - \pi$$

29)

$\frac{dy}{dx} = 1 - \frac{\pi}{2}\sec^2(\frac{\pi}{4}x)$. So $\frac{dy}{dx}\Big|_{x=4} = 1 - \frac{\pi}{2} \times \frac{1}{\cos^2(\frac{\pi}{4}x)} = 1 - \frac{\pi}{2}$. $x = 4$, $y = 4 - 2\tan(\pi) = 4$. $y - 4 = \frac{1}{\frac{\pi}{2}-1}(x - 4)$. When $y = 0$, $-4 = \frac{1}{\frac{\pi}{2}-1}(x - 4)$. $\Rightarrow -4 \times \frac{\pi}{2} + 4 = x - 4$. $\Rightarrow -2\pi + 8 = x$ so $x = 8 - 2\pi$.

30 a)

$\frac{dy}{dx} = 21\cos 7x + 4\sec^2 2x$.

30 b)

$\frac{dy}{dx} = \ln 15 \times 15^x + \frac{15}{2}\sin(\frac{5}{2}x)$.

30 c)

$\frac{dy}{dx} = \frac{1}{x} + \sqrt{2}\sec^2(\sqrt{2}x)$.

30 d)

$\frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}} - 6\cos(2x)$.

30 e)

$\frac{dy}{dx} = 3e^{3x} + 8\cos 8x$.

30 f)

$\frac{dy}{dx} = -\frac{1}{4}\sin(\frac{1}{4}x) + 2\pi\sin(2\pi x)$.

30 g)

$\frac{dy}{dx} = \ln 4 \times 4^x + \pi\sec^2(\pi x)$.

31 a)

$\frac{dy}{dx} = 15x^4 - 5e^{5x}$, $\frac{d^2y}{dx^2} = 60x^3 - 25e^{5x}$.

31 b)

$\frac{dy}{dx} = 2x - 2\cos(2x)$, $\frac{d^2y}{dx^2} = 2 + 4\sin(2x)$.

31 c)

$\frac{dy}{dx} = \frac{1}{2}x^{-\frac{1}{2}} + \frac{1}{x}$, $\frac{d^2y}{dx^2} = -\frac{1}{4}x^{-\frac{3}{2}} - x^{-2}$.

31 d)

$\frac{dy}{dx} = -x^{-2} + \frac{1}{2}\sin(\frac{1}{2}x)$. $\frac{d^2y}{dx^2} = 2x^{-3} + \frac{1}{4}\cos(\frac{1}{2}x)$.

31 e)

$\frac{dy}{dx} = 2e^{2x} - \pi\cos\pi x$. $\frac{d^2y}{dx^2} = 4e^{2x} + \pi^2\sin\pi x$.

31 f)

$\frac{dy}{dx} = -\frac{1}{2}e^{-\frac{1}{2}x} + 3\cos 3x + 3\sin 3x$. $\frac{d^2y}{dx^2} = \frac{1}{4}e^{-\frac{1}{2}x} - 9\sin 3x + 9\cos 3x$.

31 g)

$\frac{dy}{dx} = -\frac{1}{2}x^{-\frac{3}{2}} - \frac{1}{x}$. $\frac{d^2y}{dx^2} = \frac{3}{4}x^{-\frac{5}{2}} + x^{-2}$.

32 a)

$f'(x) = 2e^{2x} + \frac{1}{4}x^{-\frac{3}{4}}$ so $f'(1) = 2e^2 + \frac{1}{4}$.

32 b)

$f'(x) = \frac{5}{x} - 2x^{-3}$ so $f'(2) = \frac{5}{2} - \frac{1}{4} = \frac{9}{4}$.

32 c)

$f'(x) = \ln 2 \times 2^x + 2\pi\cos(2\pi x)$ so $f'(\frac{1}{2}) = \sqrt{2}\ln 2 - 2\pi$.

32 d)

$f'(x) = -\frac{\pi}{2}\sin(\frac{\pi}{2}x) - \frac{\pi}{2}\sec^2(-\frac{\pi}{2}x)$ so $f'(0) = -\frac{\pi}{2}$.

33)

$\frac{dy}{dx} = -2e^{-2x} - 15\cos 5x$. So $\left.\frac{dy}{dx}\right|_{x=\pi} = -2e^{-2\pi} + 15 > 0$ so increasing.

34)

$\frac{dy}{dx} = \frac{1}{x} + \pi\cos(\pi x)$ so $\left.\frac{dy}{dx}\right|_{x=3} = \frac{1}{3} - \pi$. When $x = 3$, $y = \ln 6$. $y - \ln 6 = (\frac{1}{3} - \pi)(x - 3)$. $\Rightarrow y = (\frac{1}{3} - \pi)x - 1 + 3\pi + \ln 6$.

35 a)

$\frac{dy}{dx} = -2e^{-2x} - 3\pi\sin(\pi x)$ so $\left.\frac{dy}{dx}\right|_{x=0} = -2$. When $x = 0$, $y = 1 + 3 = 4$. Tangent line: $y - 4 = -2(x) \Rightarrow y = -2x + 4$.

35 b)

Normal line: $y - 4 = \frac{1}{2}(x) \Rightarrow y = \frac{1}{2}x + 4$.

35 c)

Tangent crosses x-axis at $x = 2$. Normal crosses y-axis at $x = -8$. Therefore area of triangle is $\frac{1}{2}(2 + 8) \times 4 = 20$ units^2.

36 a)

$\frac{dy}{dx} = -3\sin(3x) + \ln 2 \times 2^x$. $\frac{d^2y}{dx^2} = -9\cos(3x) + (\ln 2)^2 \times 2^x$.

$$\left.\frac{dy}{dx}\right|_{x=\pi} = -9\cos(3\pi) + (\ln 2)^2 \times 2^\pi$$

$= 13.239987\ldots > 0$ so convex at $x = \pi$.

36 b)

$\left.\frac{dy}{dx}\right|_{x=\frac{2\pi}{3}} = -9\cos(2\pi) + (\ln 2)^2 \times 2^{\frac{2\pi}{3}}. = -6.948239\ldots < 0$ so concave at $x = \frac{2\pi}{3}$.

37)

$\frac{dy}{dx} = 3\cos 3x + 3\sec^2 3x$ stat pt when $\frac{dy}{dx} = 0$. $3\cos 3x + 3\sec^2 3x = 0 \Rightarrow \cos 3x + \frac{1}{\cos^2 3x} = 0$. $\Rightarrow \cos^3 3x + 1 = 0 \Rightarrow \cos^3 3x = -1$. $\Rightarrow \cos 3x = -1 \Rightarrow \pi = 3x \Rightarrow x = \frac{\pi}{3}$. So $(\frac{\pi}{3}, 0)$.